

Jaime ManríquezRodríguez

Applied Mathematics Ph.D.

Aspiring Computational Scientist & Simulation Engineer

My Contact

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📍 Malmö, Sweden

🌐 jaimemanriquezr

📅 Available September 2026

Competencies

- MATLAB (expert); Python, Julia (proficient); C++ (basic)
- Finite element methods, multiphysics coupling
- HPC clusters, batch workflows, scaling of long-running simulations
- Git, Unix shell, build systems, remote workflows, $\LaTeX$

Skills

- Mathematical modelling
- Interdisciplinary collaboration
- Technical writing
- Scientific communication
- Teaching and supervision

Languages

Spanish: Native 
English: Fluent 
Swedish: Conversational 

Education

Ph.D. in Applied Mathematics

LTH, Lund University | 2021 – 2026

Mathematical Civil Engineering

Universidad de Concepción | 2018 – 2020

Premio Universidad de Concepción 2020, top graduating student

BSc. engineering, mathematics

Universidad de Concepción | 2015 – 2017

About Me

I am an applied mathematician with experience in numerical simulation and multiphysics modelling. I enjoy working across disciplines and making technical ideas understandable to people with different backgrounds. Alongside my research, I have several years of experience teaching, supervising students and presenting scientific work. I am looking for technical work where I can combine problem solving, modelling and programming with collaboration and communication.

Experience

Researcher

Centre for Mathematical Sciences, Lund University | 2021 – 2026

- Built a **finite element library** for multiphysics on unfitted meshes
- Developed a **mathematical model** for slow sand filtration, predicting biofilm growth and pathogen removal
- Worked in an **interdisciplinary team** involving microbiologists and water engineers
- Taught exercise sessions at undergraduate and masters level
- Supervised students in mathematical modelling projects

Teaching Assistant

Universidad de Concepción | 2016 – 2020

- Taught numerical methods, analysis, optimisation and fluid mechanics to engineering students
- Produced video material during the pandemic

Student Internships

Chile | 2019 – 2020

- **ATCAS S.A. (2020):** Predictive model of forest-fire front propagation from terrain and weather data
- **CADETECH (2019):** Non-linear optimisation of mooring-chain placement on water vessels

Other Experiences

Publications & Dissemination

- Four peer-reviewed journal papers in computational mechanics and numerical analysis (CMAME, IMA J. Numer. Anal., Appl. Math. Modelling, CAMWA); one under review
- Seven international conference presentations, including as invited speaker
- Public presentation to high school teachers about research-level mathematics

Leadership & Academic Engagement

- Organised the department's Ph.D. seminar series; elected member of the Ph.D. student council and departmental board representative
- Undergraduate student representative at the Faculty of Engineering and the Faculty of Physical and Mathematical Sciences